Asad Amad Sheikh *Backend Developer*

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Backend engineer experienced in designing async-first, database-driven systems that prioritize performance, security, and maintainability. Proven ability to translate backend logic into reliable, production-ready APIs under real operational environments.

PROFESSIONAL EXPERIENCE

Software Engineering Intern, 06/2025 – 08/2025
Frontier Works Organisation (July -Aug 2025) Kashmir Road Rawalpindi

- Integrated internal relational databases with external Flotilla IoT platform, enabling secure operational data exchange.
- Designed and validated backend data pipelines for synchronized dashboard reporting.
- Optimized SQL queries and improved database connection handling for reliability across environments.
- Supported deployment validation, logging configuration, and backend performance checks in a high-compliance environment

Database Intern, *UFone (PTCL Group)* 06/2024 – 08/2024
Ufone Tower Blue Area Islamabad

- Queried and structured SME operational data to support enterprise reporting and decision workflows.
- Assisted in improving data consistency and resolving integrity issues within internal relational databases.
- Worked within production-scale telecom infrastructure under compliance-driven data handling practices.

SKILLS

Backend Development: Python (Production-level) | FastAPI (Async APIs, Dependency Injection, Background Tasks) | RESTful API Design (Stateless, Resource-Oriented Architecture) | Authentication & Authorization (JWT, RBAC) | Concurrency & Async Programming (async/await, task handling) | Input Validation & Data Modeling (Pydantic v2)

Databases: Databases & Data Layer | PostgreSQL (Indexing, Query Optimization, Relational Modeling) | SQL (Joins, Constraints, Transactions) | SQLAlchemy (ORM & Core) | Redis (Caching, Lightweight Messaging)

System Design & Architecture: Modular Service Architecture | API Versioning & Error Handling Standards | Logging & Rate Limiting | Clean Code & Separation of Concerns

Deployment & Infrastructure: Docker & Docker Compose | Linux CLI | Nginx + Gunicorn | CI/CD (GitHub Actions) | AWS (EC2, RDS, S3 – Practical Exposure)

Programming Languages: Python, C/C++/C#, Java, JavaScript (Node.js, Express.js), HTML/CSS.

PROJECTS

Backend Risk & Alert Management API (FastAPI), *FastAPI, Python, PostgreSQL, Redis, Docker, JWT*

- Designed and deployed asynchronous REST APIs to manage risk records, alert workflows, and evidence tracking.
- Implemented non-blocking background risk scoring pipelines using async processing and Redis-backed task handling.
- Engineered secure authentication with JWT, request throttling, and structured logging for traceability.
- Modeled relational schemas in PostgreSQL with indexing strategies to support concurrent access and low-latency queries.
- Containerized services with Docker and integrated CI validation for consistent deployment environments.

Traffic Simulator, *Asynchronous Traffic Management System — C / OS Concepts / Concurrency*

- Built a multi-threaded traffic simulation using semaphores and mutexes to enforce deadlock-free intersection control.
- Designed concurrency-safe synchronization mechanisms under resource contention scenarios.
- Applied operating system-level concurrency principles later extended to async backend task design.

Electiview – Academic Decision Support Backend, *Java · Relational Modeling · Modular Architecture*

- Engineered a modular backend system to manage electives, faculty metadata, and review workflows.
- Designed normalized relational schemas supporting filtering, ranking, and recommendation logic.
- Implemented service-layer separation to ensure maintainability and extensibility.

Autonomous AI Agent for TORCS Racing Game,
AI-Driven Autonomous Racing Agent — Python, Reinforcement Learning, REST API Integration

- Built modular RL agent integrated with external simulator APIs.
- Implemented async telemetry calls and reward-based optimization loops.

Hotel Management System, *Hotel Management Backend API — Java / SQL / File Handling*
Refactored legacy file-handling logic into modular REST-style API endpoints (mocked in Django).Implemented role-based access, authentication logic, and structured data models similar to PostgreSQL schemas.

EDUCATION

Bachelors of Science in Computer Science, *FAST NUCES* 2022 – 2026

CERTIFICATES

Ultimate Guide to FastAPI and Backend Development — Packt
Covered RESTful API development with FastAPI, async request handling, authentication (JWT), database integration, and production-oriented backend design.

Certificate of Appreciation — Frontier Works Organisation (Aug 2025)
Awarded in recognition of outstanding technical contribution during the FWO Internship.

SOFT SKILLS

Professional Strength:

- Strong ownership mindset with consistent on-time and ahead-of-deadline delivery.
- Rapid learner with ability to independently adopt new technologies and apply them in production-level projects.
- Clear and structured communicator, capable of translating technical concepts into actionable discussions.
- Collaborative team contributor experienced in working within cross-functional and compliance-driven environments.
- Disciplined work ethic with focus on reliability, consistency, and continuous improvement.